

Multistep Equations with Rational Coefficients

Clearing fractions, distributing a fractional factor, and collecting the variable on one side

The pattern to copy

- Multiplying *both* sides by a common denominator clears every fraction at once — do it before anything else, and multiply every term, including the ones with no fraction.
 - A fraction in front of a bracket still has to reach both terms inside.
 - Once the fractions are gone, collect the variable on one side and the numbers on the other, then divide.
 - Check every answer by substituting it back into the *original* equation.
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1. Solve: $\frac{x}{4} + 3 = 10$

Answer: _____

2. Solve: $\frac{2}{3}(x - 9) = 4$

Answer: _____

3. Solve: $\frac{3x}{4} = \frac{x}{2} + 5$

Answer: _____

4. Solve: $\frac{x}{2} - \frac{x}{6} = 4$

Answer: _____

5. Solve: $0.4(x + 5) = 6$

Answer: _____

6. Solve: $\frac{2x-1}{5} = 3$

Answer: _____

7. Solve: $\frac{3x}{4} - 2 = \frac{x}{6} + 12$

Answer: _____

8. Solve: $\frac{8x+12}{4} = x + 9$

Answer: _____

9. Solve: $\frac{x+2}{3} - \frac{x-4}{6} = 2$

Answer: _____

10. Solve: $0.25(x-8) = 0.5x - 7$

Answer: _____

11. Solve: $\frac{2}{5}(x+10) - \frac{x-6}{2} = 4$

Answer: _____

12. Consider the equation $\frac{3x-1}{2} + \frac{x}{4} = \frac{7x}{4} + 1$

(a) Solve it, or write down what its solution set is.

Answer: _____

(b) Explain what the last line of your working tells you about this equation.

Answer: _____